An AI-Generated Candidate Solution to the Steiner–Wiener 4-Index Inverse Problem for Finite Trees

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Released for independent mathematical verification

Abstract

For a finite tree T, the Steiner–Wiener 4-index is the sum, over all four-element vertex sets, of the number of edges in the minimal subtree spanning the set. This manuscript presents a candidate determination of which positive integers fail to occur as such an index. The candidate proof combines an exact rooted-forest dynamic program covering all trees in the exceptional range, exact caterpillar subset-sum and compressed-interval certificates for a finite bridge, and a universal Thue–Morse caterpillar construction whose fixed-order intervals overlap for all sufficiently large even orders. Every finite computation uses exact integer arithmetic and is accompanied by reproducible source, exact output, and independent checks. With $\mathbb{N} = \{1, 2, 3, \ldots\}$, the final result is

$$|\mathbb{N} \setminus \mathcal{T}_4| = 14{,}099, \qquad \max(\mathbb{N} \setminus \mathcal{T}_4) = 80{,}163.$$

Provenance and verification status

The mathematical derivations, proof construction, manuscript text, source code, computational certificates, and internal checks in this release were generated through extended ChatGPT sessions initiated by Shadow. Shadow curated, archived, and publicly released the resulting materials, but does not claim mathematical authorship and has not independently verified the proof. No independent human expert review has yet been completed.

This manuscript is therefore released as an unverified candidate solution. Its purpose is to provide a public, reproducible record and invite qualified mathematicians to examine, reproduce, confirm, correct, or refute the claimed result. The complete public archive is available at https://github.com/ShadowUR0/shadow-ai-sw4-candidate-solution.

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1 Introduction and main theorem

For a finite tree T and a nonempty set $S \subseteq V(T)$, let $d_T(S)$ denote the number of edges in the unique minimal connected subgraph of T containing S. The Steiner–Wiener 4-index is

$$\mathrm{SW}_4(T) = \sum_{\substack{S \subseteq V(T) \\ |S|=4}} d_T(S). \tag{1}$$

We write

$$\mathcal{T}_4 = \{\mathrm{SW}_4(T) : T \text{ is a finite tree}\}, \qquad \mathbb{N} = \{1, 2, 3, \ldots\}.$$

The two inverse parameters are

$$e(4) = |\mathbb{N} \setminus \mathcal{T}_4| , \qquad N_0(4) = \max(\mathbb{N} \setminus \mathcal{T}_4) ,$$

where the theorem below proves in particular that the complement is finite and nonempty.

Theorem 1.1 (Main theorem)**.** *Exactly* 14,099 *positive integers are not equal to* $\mathrm{SW}_4(T)$ *for any finite tree* T*, and the largest such integer is* 80,163*. Equivalently,*

$$\boxed{e(4) = 14{,}099}, \qquad \boxed{N_0(4) = 80{,}163}.$$

The proof has three logically distinct parts.

(i) An exact dynamic program enumerates the attainable values of *all* trees in the range in which exceptions can occur. It proves that exactly 14,099 values at most 80,163 are absent, that 80,163 is absent, and that every value from 80,164 through 178,857 occurs.

(ii) Exact caterpillar certificates and compressed interval unions prove

$$[178{,}858, 554{,}860{,}689{,}583] \subseteq \mathcal{T}_4.$$

(iii) A self-contained caterpillar construction gives

$$[272{,}308{,}250{,}181, \infty] \subseteq \mathcal{T}_4.$$

This overlaps the finite bridge.

The universal arguments are proved symbolically. Statements restricted to finite parameter ranges are explicitly identified as exact computer-assisted certificates.

2 Edge decomposition and caterpillar coordinates

We use the convention $\binom{m}{r} = 0$ for integers $m < r$.

2.1 The edge-contribution formula

Lemma 2.1 (Edge decomposition)**.** *Let* T *be a tree of order* n*. If deleting an edge* e *produces components of orders* s_e *and* $n - s_e$*, then*

$$\mathrm{SW}_4(T) = \sum_{e \in E(T)} w_n(s_e), \qquad w_n(s) = \binom{n}{4} - \binom{s}{4} - \binom{n-s}{4}. \tag{2}$$

Proof. For a four-set S, the edge e belongs to the minimal subtree spanning S if and only if S is not contained entirely in either component of $T - e$. Among all $\binom{n}{4}$ four-sets, exactly $\binom{s_e}{4}$ lie in one component and $\binom{n-s_e}{4}$ lie in the other. Thus e is counted $w_n(s_e)$ times in (1). Summing the edge indicators first and then the four-sets proves (2). □

The first difference is

$$w_n(s+1) - w_n(s) = \binom{n-s-1}{3} - \binom{s}{3}. \tag{3}$$

It is nonnegative for $1 \le s < n/2$. Choosing s_e as the smaller component order therefore gives $w_n(s_e) \ge w_n(1) = \binom{n-1}{3}$.

Corollary 2.2 (Minimum at a fixed order)**.** *The minimum Steiner–Wiener* 4*-index among trees of order* $n \geq 4$ *is*

$$G(n) = (n-1)\binom{n-1}{3}. \tag{4}$$

For $n \geq 5$*, equality holds only for the star. At* $n = 4$*, every tree has index* $3 = G(4)$*.*

Proof. Every n-vertex tree has $n-1$ edges, so the preceding inequality gives $\mathrm{SW}_4(T) \geq (n-1)w_n(1) = G(n)$. A star attains equality. For $n \geq 5$ the inequality in (3) is strict at every $1 \leq s < n/2$ once $s > 1$; hence equality forces every edge to have a one-vertex component, which characterizes the star. The order-four statement follows directly from (2). □

2.2 Caterpillar cut sets

The finite bridge and the infinite construction use a convenient exact parameterization of caterpillars.

Lemma 2.3 (Caterpillar cut-set realization)**.** *Let* $n \geq 4$ *and let* $A = \{a_1 < \cdots < a_t\} \subseteq \{2, \ldots, n-2\}$*. There is an* n*-vertex caterpillar whose non-leaf edge cuts have component orders* $a_1, \ldots, a_t$*. Its index is*

$$\mathrm{SW}_4(T_A) = G(n) + \sum_{a \in A} \Delta_n(a), \qquad \Delta_n(a) = \binom{n-1}{4} - \binom{a}{4} - \binom{n-a}{4}. \tag{5}$$

The same formula with an empty sum is realized by the star when $A = \varnothing$*.*

Proof. For $t \geq 1$, take a spine $v_0 v_1 \cdots v_t$. Attach $a_1 - 1$ leaves to v_0, attach $a_{i+1} - a_i - 1$ leaves to v_i for $1 \leq i < t$, and attach $n - a_t - 1$ leaves to v_t. All multiplicities are nonnegative. Their sum is $n - t - 1$, so the resulting tree has n vertices. Deleting the spine edge $v_{i-1}v_i$ leaves a prefix component of order a_i. Every other edge is a leaf edge.

There are $n - 1 - t$ leaf edges and t spine edges. Since

$$w_n(a) - w_n(1) = \binom{n-1}{4} - \binom{a}{4} - \binom{n-a}{4} = \Delta_n(a),$$

subtracting the star contribution and adding the t spine corrections gives (5). □

Consequently, for fixed n, every subset sum of $\Delta_n(2), \ldots, \Delta_n(n-2)$ is realized by a concrete caterpillar. This statement is used repeatedly below; it is not merely a numerical encoding of cut sizes.

3 The exact all-tree certificate

This section describes the finite computation that determines every exception. Its completeness rests on a rooted-tree recurrence, not on a database of selected tree shapes.

For subsets $X, Y \subseteq \mathbb{N} \cup \{0\}$, write $X + Y = \{x + y : x \in X,\ y \in Y\}$. Fix a global tree order n. Let $P_n(r)$ be the set of internal edge sums of rooted trees of order r, where the edge from the root to a possible parent is omitted and every internal edge is evaluated with the global weight w_n. Let $F_n(t)$ be the corresponding set for ordered forests of rooted trees with total order t.

Proposition 3.1 (Rooted-forest recurrence)**.** *For fixed* n*,*

$$F_n(0) = \{0\}, \qquad\qquad P_n(1) = \{0\}, \tag{6}$$

$$P_n(r) = F_n(r-1) \qquad (r \geq 2), \tag{7}$$

$$F_n(t) = \bigcup_{s=1}^{t} \left(F_n(t-s) + w_n(s) + P_n(s)\right) \qquad (t \geq 1). \tag{8}$$

Moreover, $P_n(n)$ is exactly the set of values $\mathrm{SW}_4(T)$ over all n-vertex trees T.

Proof. A rooted tree of order r consists of its root together with a forest of rooted child components of total order $r-1$, proving (7). For an ordered forest of total order t, choose its first rooted component to have order s. Its attachment edge contributes $w_n(s)$ and its internal edges contribute an element of $P_n(s)$; the remaining ordered forest contributes an element of $F_n(t-s)$. This proves that every construction on the right side of (8) is valid.

Conversely, every rooted forest may be ordered arbitrarily, and after choosing its first component it appears in one of the terms on the right. Thus the recurrence omits no rooted forest. Ordering children may duplicate a value, but creates no invalid value. Rooting any unrooted n-vertex tree at an arbitrary vertex does not change its edge contributions, so $P_n(n)$ is the exact all-tree value set. □

3.1 Bitset implementation and finite cutoffs

For a cap B, a set $X \subseteq \{0, \ldots, B\}$ is encoded by the Python integer

$$\beta(X) = \sum_{x \in X} 2^x.$$

A shift by c represents $X + c$. The Minkowski sum $X + Y$ is computed exactly by iterating over the set bits of the smaller operand and OR-ing shifted copies of the other operand. Bits above B are discarded. Since every edge weight is nonnegative, discarding a partial sum above B cannot remove a later sum at most B.

Computer-assisted certificate 3.2 (Exceptional range)**.** The program `sw4_certificate.py`, implementing Proposition 3.1 with arbitrary-precision Python integers, certifies the following statements.

(a) Exactly 14,099 positive integers at most 80,163 do not belong to $\mathcal{T}_4$.
(b) The integer 80,163 does not belong to $\mathcal{T}_4$.
(c) Every integer in $[80{,}164, 178{,}857]$ belongs to $\mathcal{T}_4$.

Why the finite computation is exhaustive. For the first two assertions, Corollary 2.2 gives

$$G(29) = 91{,}728 > 80{,}163.$$

Hence no tree of order at least 29 can contribute a value in the exceptional range. The program computes $P_n(n) \cap [0, 80{,}163]$ exactly for every $4 \leq n \leq 28$ and takes their union.

For the third assertion,

$$G(34) = 180{,}048 > 178{,}857,$$

so it is enough to compute the exact sets for $4 \leq n \leq 33$ up to the cap 178,857. The program asserts that the missing list up to this larger cap is identical to the 14,099-element list from the first computation. The list is stored in `sw4_missing_values.txt`; the maximal consecutive ranges are stored in `sw4_missing_ranges.txt`. □

The same source also performs an exact caterpillar subset-sum computation. Since

$$G(81) = 6{,}572{,}800,$$

orders at most 80 suffice below $G(81)$. Applying Lemma 2.3 to all subsets of $\{2, \ldots, n-2\}$ for $4 \le n \le 80$ gives the next certificate.

Computer-assisted certificate 3.3 (Initial caterpillar interval)**.** Every integer in

$$[178{,}858, 6{,}572{,}799]$$

belongs to $\mathcal{T}_4$.

For fixed n, the subset sums are encoded by the exact update

$$B \longleftarrow B \text{ OR } (B \ll \Delta_n(a)), \qquad a = 2, \ldots, n-2,$$

with a mask at the stated cap. This enumerates every subset and no other choice.

4 Central moment switches

We next develop the common algebra behind the finite compressed bridge and the universal tail.

4.1 Even-order coordinates

Put $n = 2h$ and define, for $0 \le x \le h-2$,

$$\delta_h(x) = \Delta_{2h}(h + x). \tag{9}$$

The distance $x = 0$ corresponds to the single cut size h. Every positive x corresponds to two distinct cut sizes $h - x$ and $h + x$; because $\Delta_{2h}(h - x) = \Delta_{2h}(h + x)$, these are two labeled slots of the same weight $\delta_h(x)$.

Expanding (5) gives

$$\begin{aligned} \delta_h(x) = \; & -\frac{x^4}{12} + \left(-\frac{h^2}{2} + \frac{3h}{2} - \frac{11}{12}\right) x^2 \\ & + \frac{7h^4}{12} - \frac{17h^3}{6} + \frac{59h^2}{12} - \frac{11h}{3} + 1. \end{aligned} \tag{10}$$

Thus δ_h is an even quartic in x.

For a finite multiset S of nonnegative distances, write

$$\Phi_h(S) = \sum_{x \in S} \delta_h(x),$$

with multiplicities.

Lemma 4.1 (Moment switch)**.** *Let P, Q be finite multisets satisfying*

$$|P| = |Q|, \qquad \sum_{x \in P} x^2 = \sum_{x \in Q} x^2.$$

Then

$$\Phi_h(P) - \Phi_h(Q) = -\frac{1}{12} \left(\sum_{x \in P} x^4 - \sum_{x \in Q} x^4 \right), \tag{11}$$

which is independent of h.

Proof. In (10), equal cardinalities cancel the constant term and equal square sums cancel the coefficient of x^2. The displayed fourth-power term is the only term that remains. $\square$

A *state group* is a finite collection of multisets satisfying the two common-moment conditions in Lemma 4.1. Its normalized increments are its state values minus their minimum.

Lemma 4.2 (Independent slot allocation)**.** *For a state group $\mathcal{S}$, let*

$$r_{\mathcal{S}}(x) = \max_{S\in\mathcal{S}} m_S(x),$$

where $m_S(x)$ is the multiplicity of x in S. Suppose a collection of state groups satisfies

$$\sum_{\mathcal{S}} r_{\mathcal{S}}(0) \le 1, \qquad \sum_{\mathcal{S}} r_{\mathcal{S}}(x) \le 2 \quad (x > 0).$$

Then all group choices can be made independently in a single caterpillar. Every labeled slot not reserved by a group may also be made independently optional.

Proof. Label the one slot at distance 0 and the two slots at each positive distance. For each group and distance, reserve $r_{\mathcal{S}}(x)$ labeled slots, with reservations disjoint across groups. Every state uses no more than its reservation. A chosen collection of states therefore selects a set of distinct cut sizes. Lemma 2.3 realizes their union as a caterpillar. Unreserved slots are disjoint from every state and may be selected or omitted independently. $\square$

Lemma 4.3 (Bounded complete sequence)**.** *Suppose independent choices already realize every integer in $[0, W]$. If an additional independent binary switch has increment $d \le W + 1$, then the enlarged choices realize every integer in $[0, W + d]$. More generally, A independent copies of that switch extend the interval to $[0, W + Ad]$.*

Proof. One copy gives the union $[0, W] \cup [d, d + W]$. The hypothesis makes these intervals overlap or touch. Repeating the same argument A times proves the second statement. $\square$

4.2 The universal and full seeds

The first state group is

$$\begin{aligned} S_0 &= (0, 2, 2, 3, 6, 6, 10, 10, 12, 12), \\ S_1 &= (1, 1, 4, 4, 6, 7, 8, 9, 12, 13), \\ S_2 &= (1, 1, 4, 5, 5, 7, 7, 11, 11, 13). \end{aligned}$$

All three have cardinality 10 and square sum 577; their fourth-power sums are 64,177, 64,165, and 64,153. By Lemma 4.1, their normalized increments are $0, 1, 2$. The independent pair

$$\begin{aligned} X &= (16, 22, 24, 27, 30, 32, 36), \\ Y &= (17, 20, 25, 26, 31, 33, 35) \end{aligned}$$

has cardinality 7, square sum 5,265, and fourth-power sums 4,701,201 and 4,701,189, so it has increment 1. These two groups realize $[0, 3]$.

Appendix A lists fifteen further binary pairs with increments

$$4, 8, 16, \ldots, 65{,}536.$$

Their simultaneous demand is at most one at distance zero and at most two at every positive distance; Appendix B records the compact capacity audit. Successively applying Lemma 4.3 gives the universal seed interval

$$[0, 131{,}071]. \tag{12}$$

Its largest distance is 99.

For finite junction calculations only, one more pair is used:

$$(32, 52, 76, 100), \qquad (36, 44, 84, 96).$$

Both states have square sum 19,504, and their fourth-power sums are 141,722,368 and 140,149,504, giving increment 131,072. Together with (12), this produces the full seed

$$[0, 262{,}143]. \tag{13}$$

Its largest distance is 100, and its full simultaneous capacity is valid.

5 Exact finite bridges

5.1 Bitset bridge through 292,031,545

For each $70 \le n \le 105$, the program `sw4_caterpillar_bridge_certificate.py` computes every bounded subset sum of $\Delta_n(2), \ldots, \Delta_n(n-2)$, shifts by $G(n)$, and ORs the results over all orders. The cap is 292,031,545.

Computer-assisted certificate 5.1 (Lower caterpillar bridge)**.** The exact bit mask contains every bit in

$$[6{,}572{,}800, 292{,}031{,}545]\,.$$

Therefore this interval is contained in $\mathcal{T}_4$.

The update is the standard exact subset recurrence from Certificate 3.3. There is no sampling, hashing, modular filter, or floating-point arithmetic.

5.2 Exact compressed interval unions

The next certificate uses the fact that a finite subset of $\mathbb{Z}$ can be stored exactly as a union of maximal integer intervals.

Lemma 5.2 (Exact optional-weight update)**.** *Let $S \subseteq \mathbb{Z}$ be represented exactly by pairwise-disjoint maximal integer intervals. For an optional weight $w \ge 0$, the attainable set after allowing the choices 0 and w is*

$$S' = S \cup (S + w).$$

Merging the two sorted interval lists and coalescing only overlapping or adjacent intervals produces the exact maximal-interval representation of S'.

Proof. Translation sends each interval $[a, b]$ of S to $[a + w, b + w]$ and no other integer. The union of the original and translated lists is therefore exactly S'. Coalescing overlapping or adjacent intervals changes only its representation, not the represented set. $\square$

For a fixed half-order h, the certificate constructs a baseline consisting of $G(2h)$ plus the minimum value from every reserved state group. The seed state choices give an initial interval $[0, W_{\mathrm{seed}}]$. Every annulus switch and every remaining labeled cut slot supplies an independent optional weight. Lemma 5.2 is applied to every weight. Thus the final interval union is the exact attainable increment set for the stated caterpillar family.

Three seed families are used. Their complete states are listed in Appendix C; the certificate checks equal moments, normalized increments, support validity, and simultaneous capacities before processing any order.

Table 1: Families used in the compressed finite bridge. The displayed first and last intervals are the selected longest components at the endpoints of each order range.

Family	h range	Seed width	Fixed annuli	Endpoint intervals
$\mathcal{F}_{51}$	63–66	63	none	$[282{,}204{,}658, 495{,}014{,}629]$; $[346{,}215{,}511, 635{,}465{,}903]$
$\mathcal{F}_{65}$	67–206	2,047	none	$[494{,}227{,}243, 564{,}426{,}072]$; $[58{,}222{,}371{,}640, 236{,}605{,}611{,}082]$
$\mathcal{F}_{100}$	206–253	262,143	$(101, 5), (141, 8)$	$[106{,}740{,}323{,}728, 189{,}108{,}214{,}894]$; $[272{,}308{,}250{,}181, 554{,}860{,}689{,}583]$

The full output contains 192 order rows. Each selected interval begins no later than one past the previously covered endpoint. The first interval lies inside Certificate 5.1; every family transition also overlaps.

Computer-assisted certificate 5.3 (Compressed finite bridge)**.** The program `sw4_complete_bridge_certificate.cpp` and its exact output `sw4_complete_bridge_output.txt` certify

$$[282{,}204{,}658, 554{,}860{,}689{,}583] \subseteq \mathcal{T}_4.$$

Together with Certificate 5.1,

$$\boxed{[6{,}572{,}800, 554{,}860{,}689{,}583] \subseteq \mathcal{T}_4.} \tag{14}$$

The independent Python checker `sw4_bridge_independent_check.py` reads all 192 rows, verifies every junction, and recomputes the six boundary orders independently.

Remark 5.4 (Necessary correction to a weaker propagation argument). It is essential that the certificate use the exact interval update of Lemma 5.2. At $h = 206$, after a partial insertion, a raw weight 12,794,200 exceeds the then-current complete width plus one, 12,772,104. Hence the sufficient condition of Lemma 4.3 does not propagate a single interval through that ordering. The exact attainable set temporarily has more than one component; subsequent weights reconnect the components. The compressed interval algorithm retains every component and therefore proves the stated interval without the invalid stronger claim.

6 Thue–Morse annuli

We now prove the universal infinite tail. The construction uses the universal seed (12), not the extra 131,072 digit.

6.1 An eight-point identity

For integers $R, q \geq 1$ and $0 \leq r < q$, set

$$x_j = R + r + jq, \qquad 0 \leq j \leq 7.$$

Partition the indices according to the parity of the number of ones in the binary expansion of j. Equivalently, put $\varepsilon_j = 1$ for $j \in \{0, 3, 5, 6\}$ and $\varepsilon_j = -1$ for $j \in \{1, 2, 4, 7\}$.

A direct summation gives

$$\begin{array}{c|ccccc} m & 0 & 1 & 2 & 3 & 4 \\ \hline \sum_{j=0}^{7} \varepsilon_j j^m & 0 & 0 & 0 & -48 & -672. \end{array} \tag{15}$$

Let $E_{R,q,r}$ and $O_{R,q,r}$ denote the two four-element distance states.

Lemma 6.1 (Thue–Morse switch)**.** *The two states have an index difference independent of h, of absolute value*

$$d_{R,q,r} = 8q^3(2(R + r) + 7q). \tag{16}$$

For fixed R, q, the q switches $r = 0, \dots, q-1$ have disjoint supports, and their supports partition the annulus of distances

$$R, R+1, \dots, R + 8q - 1.$$

Their total increment is

$$D(R, q) = 8q^4(2R + 8q - 1). \tag{17}$$

Proof. Write $t = R + r$. In the signed difference of the two state values, the constant and quadratic parts of (10) vanish by (15). Expanding the fourth power gives

$$\sum_{j=0}^{7} \varepsilon_j (t + jq)^4 = 4tq^3(-48) + q^4(-672).$$

Multiplication by the coefficient $-1/12$ in (10) yields

$$16tq^3 + 56q^4 = 8q^3(2t + 7q),$$

which is (16).

Every integer in $[R, R + 8q - 1]$ has a unique expression $R + r + jq$ with $0 \leq r < q$ and $0 \leq j \leq 7$, proving the support assertion. Finally,

$$\begin{aligned} \sum_{r=0}^{q-1} d_{R,q,r} &= 8q^3 \sum_{r=0}^{q-1} (2R + 2r + 7q) \\ &= 8q^4(2R + 8q - 1). \end{aligned}$$

□

7 A universal fixed-order interval

7.1 The annulus ladder

Define

$$R_0 = 100, \qquad q_0 = 4, \qquad R_{i+1} = R_i + 8q_i, \qquad q_{i+1} = \left\lceil \frac{3q_i}{2} \right\rceil. \tag{18}$$

Let

$$W_0 = 131{,}071, \qquad W_{k+1} = W_k + D(R_k, q_k). \tag{19}$$

Thus W_k is the complete switch width after the universal seed and the first k annuli, indexed $0, \ldots, k-1$.

Lemma 7.1 (Annulus ladder completeness)**.** *For every $k \geq 0$, the universal seed together with the first k annuli realizes every increment in $[0, W_k]$.*

Proof. For the first three annuli, the first digits and preceding widths are

$$116{,}736 \leq 131{,}072, \qquad 528{,}768 \leq 604{,}160, \qquad 2{,}466{,}936 \leq 3{,}828{,}608.$$

Thus Lemma 4.3 starts the induction.

We first prove the invariant

$$12q_i \leq R_i \leq 25q_i. \tag{20}$$

It holds for $(R_0, q_0) = (100, 4)$. Since $q_i \geq 4$,

$$\frac{3}{2}q_i \leq q_{i+1} \leq \frac{13}{8}q_i.$$

Consequently,

$$R_{i+1} \leq 33q_i \leq 22q_{i+1} < 25q_{i+1},$$

and

$$R_{i+1} \geq 20q_i \geq \frac{160}{13}q_{i+1} > 12q_{i+1}.$$

Consider an annulus with index $i \geq 3$. Put $p = q_{i-1}$ and $R = R_{i-1}$. Then $p \geq 9$, $R/p \geq 12$, and $q_i/p \leq 14/9$. Since $R_i = R + 8p$, equation (16) gives

$$\frac{d_{R_i,q_i,0}}{8p^4} \leq \left(\frac{14}{9}\right)^3 \left(2\frac{R}{p} + \frac{242}{9}\right). \tag{21}$$

The preceding annulus has total increment

$$\frac{D(R,p)}{8p^4} = p\left(2\frac{R}{p} + 8 - \frac{1}{p}\right). \tag{22}$$

The difference between the right side of (22) and the right side of (21) is increasing in R/p. Its minimum in the stated domain occurs at $p = 9$ and $R/p = 12$, where the required inequality is the exact integer comparison

$$287 \cdot 6561 > 2744 \cdot 458.$$

Therefore

$$d_{R_i,q_i,0} \leq D(R_{i-1}, q_{i-1}) \leq W_i.$$

Within a fixed annulus,

$$d_{R,q,r+1} - d_{R,q,r} = 16q^3,$$

and $d_{R,q,r+1} \leq 2d_{R,q,r}$. If the current width before inserting $d_{R,q,r}$ is at least $d_{R,q,r} - 1$, then afterward it is at least $2d_{R,q,r} - 1$, so the next digit also satisfies the hypothesis of Lemma 4.3. This proves the induction over all digits and all annuli. □

7.2 Reaching the raw caterpillar cuts

For $h \geq 365$, let $k = k(h)$ be the largest integer satisfying

$$R_k \leq h - 1. \tag{23}$$

Then the first k annuli occupy distances 100 through $R_k - 1$ and fit inside $0 \leq x \leq h - 2$. Maximality gives $h \leq R_{k+1}$. The smallest nonzero raw caterpillar increment is

$$\Delta_{2h}(2) = \binom{2h-2}{3}. \tag{24}$$

Lemma 7.2 (Switch width reaches the first raw cut)**.** *For every $h \geq 365$,*

$$W_{k(h)} \geq \Delta_{2h}(2) - 1. \tag{25}$$

Proof. The first relevant ladder values are

k	W_k	$\binom{2R_{k+1}-2}{3} - 1$
4	215,457,655	199,064,819
5	1,607,941,615	648,686,323
6	12,672,515,567	2,138,222,579

These are exact evaluations of (18)–(19).

For $k \geq 7$, put $i = k - 1$ and $p = q_i$. Then $p \geq 48$. From (20) and $q_{i+1} \leq 13p/8$,

$$R_{i+2} = R_i + 8p + 8q_{i+1} \leq 46p.$$

Moreover,

$$W_k \geq D(R_i, p) = 8p^4(2R_i + 8p - 1) \geq 64p^5.$$

On the other hand,

$$\binom{2R_{i+2}-2}{3} < \frac{(2R_{i+2})^3}{6} \leq \frac{4}{3}\, 46^3 p^3.$$

The inequality

$$64 \cdot 48^2 > \frac{4}{3}\, 46^3$$

therefore implies $W_k \geq \binom{2R_{k+1}-2}{3} - 1$. Since $h \leq R_{k+1}$, monotonicity of the binomial coefficient and (24) prove (25). $\square$

7.3 Insertion of all remaining slots

For $1 \leq a < n/2$, equation (3) implies that the first differences of $\Delta_n(a)$ are positive and nonincreasing. Since $\Delta_n(1) = 0$, we obtain

$$\Delta_n(a) \leq (a-1)\Delta_n(2), \qquad 2 \leq a \leq n/2. \tag{26}$$

The seed uses only distances at most 99. Each annulus uses one of the two slots at each supported positive distance. Therefore, for every $a = 2, \ldots, h - 100$, at least one slot of weight $\Delta_{2h}(a)$ remains unreserved: it is the slot at distance $x = h - a \geq 100$.

Starting from the complete interval $[0, W_k]$, insert one such slot in the order $a = 2, 3, \ldots, h-100$. Lemma 7.2 handles $a = 2$. After the slots through a have been inserted, the current width W satisfies

$$\begin{aligned} W + 1 &\geq \Delta_{2h}(2) + \sum_{j=2}^{a} \Delta_{2h}(j) \\ &\geq a\Delta_{2h}(2) \\ &\geq \Delta_{2h}(a+1), \end{aligned}$$

where the last line uses (26). Thus the interval remains complete.

Let

$$S_h = \sum_{a=2}^{h-100} \Delta_{2h}(a). \tag{27}$$

Hockey-stick summation gives

$$\begin{aligned} S_h = (h-101)\binom{2h-1}{4} - \binom{h-99}{5} \\ - \binom{2h-1}{5} + \binom{h+100}{5}. \end{aligned} \tag{28}$$

Furthermore, exact expansion yields

$$\begin{aligned} 120(S_h - \Delta_{2h}(h)) = 48h^5 - 7315h^4 + 34790h^3 \\ + 19641055h^2 - 59058458h + 19539439680. \end{aligned} \tag{29}$$

For $h \geq 365$, the first two terms combine to $h^4(48h - 7315) > 0$, the terms $19641055h^2 - 59058458h$ combine to a positive quantity, and every other term displayed in (29) is positive. Hence

$$S_h \geq \Delta_{2h}(h). \tag{30}$$

After the first copy at every outer size has been inserted, the accumulated width is at least S_h, hence at least the largest raw cut weight by (30). Every remaining unreserved slot may therefore be inserted in nondecreasing order using Lemma 4.3.

Define L_h explicitly as

$$\begin{aligned} L_h = G(2h) + \sum_{\mathcal{S} \in \mathcal{G}} \min_{S \in \mathcal{S}} \Phi_h(S) \\ + \sum_{i=0}^{k(h)-1} \sum_{r=0}^{q_i-1} \min\{\Phi_h(E_{R_i,q_i,r}), \Phi_h(O_{R_i,q_i,r})\}, \end{aligned} \tag{31}$$

where $\mathcal{G}$ is the universal seed. Let U_h be L_h plus the complete switch width and the weights of all remaining optional slots.

Theorem 7.3 (Universal fixed-order interval)**.** *For every integer $h \geq 365$, the caterpillars described above realize every integer in*

$$J_h = [L_h, U_h].$$

At the first order,

$$J_{365} = [1{,}596{,}122{,}112{,}921, 3{,}577{,}546{,}733{,}067]. \tag{32}$$

Proof. The seed and annulus states are simultaneously valid by Lemma 4.2; their choices realize $[0, W_{k(h)}]$ by Lemma 7.1. The preceding insertion argument adds every remaining labeled slot while preserving a complete interval. Translating by the baseline (31) proves the theorem. The endpoint evaluation at $h = 365$ is an exact integer calculation verified independently by the supplied Python certificates. □

8 Overlap of the universal intervals

8.1 Finite overlap range

The exact program `sw4_infinite_tail_certificate.py` evaluates the explicit construction of Theorem 7.3 for every $365 \le h \le 1171$. It verifies

$$L_{h+1} \le U_h + 1, \qquad 365 \le h \le 1170. \tag{33}$$

The minimum overlap margin $U_h + 1 - L_{h+1}$ is 1,960,269,091,932, at the transition $h = 365$ to 366. The 807 exact rows are stored in `sw4_infinite_tail_overlaps.csv`.

8.2 Symbolic overlap range

At order $2(h+1)$, every selected universal seed state contains exactly 77 cuts: ten from the three-state group, seven from the unit pair, and four from each of the fifteen binary groups. If $k = k(h+1)$ annuli are selected, they use $4\sum_{i<k} q_i$ cuts. From $R_k = 100 + 8\sum_{i<k} q_i$ and $R_k \le h$, the total number of selected baseline cuts is at most

$$77 + \frac{h - 100}{2} = \frac{h + 54}{2}. \tag{34}$$

Every selected cut at order $2h + 2$ contributes at most $\binom{2h+1}{4}$. Therefore

$$L_{h+1} \le G(2h + 2) + \frac{h + 54}{2}\binom{2h + 1}{4}. \tag{35}$$

On the other hand, Theorem 7.3 and the insertion of the outer cuts give

$$U_h \ge G(2h) + S_h. \tag{36}$$

Substituting (28) into (35)–(36) and clearing denominators gives

$$\begin{aligned} &120\left(G(2h) + S_h + 1 - G(2h + 2) - \frac{h + 54}{2}\binom{2h + 1}{4}\right) \\ &= 8h^5 - 9365h^4 + 35340h^3 + 19643615h^2 \\ &\quad - 59060078h + 19539440040. \end{aligned} \tag{37}$$

For $h \ge 1171$, the first two terms combine to $h^4(8h - 9365) > 0$, the pair $19643615h^2 - 59060078h$ is positive, and the remaining displayed terms are positive. Hence (37) is positive and

$$L_{h+1} \le U_h + 1, \qquad h \ge 1171. \tag{38}$$

Combining (33) and (38) proves the universal overlap theorem.

Theorem 8.1 (Universal tail above J_{365})**.**

$$\bigcup_{h\geq 365} J_h = [1{,}596{,}122{,}112{,}921, \infty]\,.$$

Proof. The first interval is (32). Every consecutive pair touches or overlaps by (33) and (38). The left endpoints tend to infinity because $L_h \geq G(2h)$, so the union is exactly the displayed ray. $\square$

9 Exact finite junction to the universal tail

The first universal interval begins above the finite bridge (14). A final exact chain uses the full seed (13), the fixed annuli

$$(R, q) = (101, 5), \qquad (R, q) = (141, 8), \tag{39}$$

and every remaining labeled cut slot. For each $h = 253, 254, \ldots, 365$, the C++ program `sw4_infinite_tail_finite_overlap.cpp` applies the exact interval update of Lemma 5.2 and selects a longest component.

The first selected interval, at $h = 253$, is

$$[272{,}308{,}250{,}181, 554{,}860{,}689{,}583], \tag{40}$$

which already lies in and reaches the end of the finite bridge. Every one of the 113 selected intervals overlaps the preceding covered set. The last, at $h = 365$, is

$$[1{,}304{,}696{,}148{,}031, 3{,}868{,}972{,}697{,}957], \tag{41}$$

which overlaps J_{365}.

Computer-assisted certificate 9.1 (Finite junction)**.** The exact interval rows in `sw4_infinite_tail_finite_overlap.csv` certify

$$[272{,}308{,}250{,}181, 3{,}868{,}972{,}697{,}957] \subseteq \mathcal{T}_4,$$

and the final endpoint overlaps the first universal interval.

Corollary 9.2 (Self-contained infinite tail)**.**

$$\boxed{[272{,}308{,}250{,}181, \infty] \subseteq \mathcal{T}_4.} \tag{42}$$

Proof. Certificate 9.1 joins continuously to Theorem 8.1. $\square$

10 Deduction of the exact inverse parameters

We now combine the logically independent pieces.

Proof of Theorem 1.1. Certificate 3.2 proves that exactly 14,099 positive integers at most 80,163 are unattainable, that 80,163 itself is unattainable, and that every integer from 80,164 through 178,857 is attainable. Certificate 3.3 continues the coverage through 6,572,799. Equation (14) continues it through 554,860,689,583. Finally, the infinite tail (42) begins below that endpoint. Hence every integer greater than 80,163 is attainable, while the certified 14,099 exceptions at or below it are the complete complement. Therefore

$$e(4) = 14{,}099, \qquad N_0(4) = 80{,}163.$$

$\square$

11 Computational model, reproducibility, and audit

11.1 Arithmetic and algorithmic completeness

All Python computations use arbitrary-precision integers. The bitset certificates therefore have no fixed-width overflow. The C++ programs use signed 128-bit intermediates for binomial products and assert before every conversion that stored endpoints fit in signed 64-bit range. No floating-point arithmetic is used.

The roles of the programs are as follows.

`sw4_certificate.py`
: Implements the exact rooted-forest recurrence and the original caterpillar subset-sum certificate. Its order bounds are justified by the star minimum, so it covers every tree order that can affect the claimed ranges.

`sw4_caterpillar_bridge_certificate.py`
: Computes all bounded caterpillar subset sums for orders 70 through 105 and checks the entire required bit mask.

`sw4_complete_bridge_certificate.cpp`
: Computes exact interval unions for the 192 finite-bridge rows and asserts every junction.

`sw4_bridge_independent_check.py`
: Independently checks all bridge rows and recomputes the six family-boundary orders.

`sw4_infinite_tail_finite_overlap.cpp`
: Computes the 113 exact finite-junction rows from $h = 253$ through 365.

`sw4_infinite_tail_certificate.py`
: Checks every seed witness and capacity, all complete-sequence inequalities in the universal construction, the explicit interval at $h = 365$, and every finite overlap through $h = 1171$.

`sw4_infinite_tail_independent_check.py`
: Independently checks the witness tables, both finite chains, all 807 universal overlap rows, and the symbolic boundary identities.

The exact optional-weight update of Lemma 5.2 is exhaustive because each step replaces the current exact set S by precisely $S \cup (S + w)$. Induction on the number of optional weights proves that the final union contains every and only the sums generated by the stated family. Likewise, the bitset subset update is the exact recurrence for choosing or omitting each cut.

11.2 Reproduction commands

From the supplementary archive directory, the principal checks are reproduced by

```
python sw4_certificate.py
python sw4_caterpillar_bridge_certificate.py

g++ -O3 -std=c++20 -Wall -Wextra -pedantic \
  sw4_complete_bridge_certificate.cpp \
  -o sw4_complete_bridge_certificate
./sw4_complete_bridge_certificate
python sw4_bridge_independent_check.py

g++ -O3 -std=c++20 -Wall -Wextra -pedantic \
  sw4_infinite_tail_finite_overlap.cpp \
  -o sw4_infinite_tail_finite_overlap
./sw4_infinite_tail_finite_overlap
python sw4_infinite_tail_certificate.py
```

```
python sw4_infinite_tail_independent_check.py

sha256sum -c sw4_self_contained_sha256.txt
```

The expected summaries include

```
PASS
candidate e(4) = 14099
candidate N0(4) = 80163
exact all-tree coverage: 80164..178857
caterpillar coverage: 178858..6572799

PASS
orders=70..105
coverage=6572800..292031545

PASS
bridge_start=282204658
bridge_end=554860689583

PASS
bridge_start=272308250181
bridge_end=3868972697957
universal_L_365=1596122112921

PASS
universal_h_start=365
universal_first_interval=1596122112921..3577546733067
symbolic_overlap_h>=1171
```

The verified supplementary archive has SHA-256

`1ed15206d460cac319b0b3683a41ade37db687ecd590a199126b08e189b47342`.

Selected file hashes appear in Appendix D; the full manifest is `sw4_self_contained_sha256.txt`. The supplementary archive is cited as reference [1].

11.3 Hostile audit summary

The following failure modes are explicitly excluded.

(1) Proposition 3.1 covers all rooted trees; the star lower bound excludes all uncomputed orders below the exceptional cutoff.
(2) Every caterpillar state is a set of labeled cut slots satisfying the one-at-zero and two-at-positive-distance capacities; Lemma 2.3 converts every selected set into an actual finite tree.
(3) Every seed row has equal cardinality and equal square sum, and its fourth-power difference gives the claimed exact increment.
(4) Thue–Morse supports are pairwise disjoint within an annulus and across the ladder; their increment formula is derived symbolically.
(5) The universal complete-sequence inequalities are proved for all annuli, not inferred from numerical density.
(6) The finite interval programs preserve every component. In particular, the invalid stronger one-interval propagation noted after Certificate 5.3 is not used.

(7) Every finite interval junction is checked with exact integers. The transition to the universal range is exact, and the symbolic overlap polynomial is positive throughout its entire claimed domain.
(8) The final finite bridge and the infinite tail overlap, leaving no unproved integer above 80,163.

A Universal central-seed witnesses

For each row below, the two states have the same cardinality and the same square sum. The normalized increment is the absolute fourth-power difference divided by 12, as in Lemma 4.1. The orientation of the states is irrelevant because the lower-valued state is used in the baseline.

Table 2: Binary witnesses for the universal central seed.

d	State P	State Q	Common $\sum x^2$	$\sum_P x^4$	$\sum_Q x^4$
4	$72, 78, 85, 90$	$73, 76, 88, 88$	26,593	181,699,537	181,699,489
8	$8, 58, 71, 73$	$45, 46, 51, 84$	13,798	65,130,514	65,130,418
16	$77, 83, 90, 95$	$78, 81, 93, 93$	29,943	229,671,987	229,672,179
32	$81, 87, 94, 99$	$82, 85, 97, 97$	32,767	274,470,979	274,471,363
64	$22, 43, 64, 71$	$25, 48, 54, 75$	11,470	45,841,954	45,842,722
128	$15, 28, 39, 51$	$23, 23, 37, 52$	5,131	9,743,923	9,745,459
256	$9, 24, 64, 87$	$29, 40, 40, 91$	12,322	74,405,314	74,402,242
512	$14, 14, 29, 35$	$18, 18, 21, 37$	2,458	2,284,738	2,278,594
1,024	$28, 31, 86, 92$	$30, 33, 80, 96$	17,605	127,878,289	127,890,577
2,048	$16, 60, 66, 66$	$16, 62, 62, 68$	12,568	50,975,008	50,999,584
4,096	$55, 61, 67, 70$	$57, 58, 69, 69$	16,135	67,157,587	67,206,739
8,192	$26, 41, 53, 65$	$27, 38, 57, 63$	9,391	29,023,843	28,925,539
16,384	$3, 60, 67, 75$	$19, 53, 68, 77$	13,723	64,751,827	64,555,219
32,768	$13, 20, 43, 47$	$15, 21, 44, 45$	4,627	8,487,043	8,093,827
65,536	$38, 50, 56, 70$	$42, 46, 54, 72$	11,980	42,179,632	42,966,064

B Capacity audit for the central seeds

For the universal seed, the total reserved demand equals one at

$$\{0, 17, 19, 32, 36, 39, 41, 42, 44, 47, 48, 50, 52, 55, 56, 61, 63, 65,\\ 76, 80, 82, 83, 84, 86, 91, 92, 94, 95, 96, 99\},$$

equals two at

$$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18, 20, 21, 22, 23, 24, 25, 26,\\ 27, 28, 29, 30, 31, 33, 35, 37, 38, 40, 43, 45, 46, 51, 53, 54, 57, 58, 60, 62, 64,\\ 66, 67, 68, 69, 70, 71, 72, 73, 75, 77, 78, 81, 85, 87, 88, 90, 93, 97\},$$

and is zero at the other distances through 99. Thus the one-slot capacity at zero and the two-slot capacity at every positive distance are respected. The full seed adds the 131,072 pair. Its exact demand remains at most one at zero and at most two elsewhere; its largest used distance is 100. The machine-readable versions are `sw4_central_seed_capacity.csv` and `sw4_full_seed_capacity.csv`.

C Compact seeds used in the finite bridge

All families include the common three-state group S_0, S_1, S_2 and the unit pair X, Y. The following tables list the remaining binary pairs. In each row the common square sum and the fourth-power difference verify the displayed increment through Lemma 4.1.

Table 3: Additional pairs for $\mathcal{F}_{51}$, yielding seed width 63.

d	P	Q	$\sum x^2$	$\sum_P x^4$	$\sum_Q x^4$
4	$23, 28, 32, 49$	$18, 25, 42, 45$	4,738	7,707,874	7,707,922
8	$16, 24, 45, 51$	$8, 28, 47, 49$	5,458	11,263,138	11,263,234
16	$20, 29, 46, 50$	$14, 36, 42, 51$	5,857	11,594,737	11,594,929
32	$9, 39, 39, 46$	$23, 29, 37, 50$	5,239	9,110,899	9,111,283

Table 4: Additional pairs for $\mathcal{F}_{65}$, yielding seed width 2,047.

d	P	Q	$\sum x^2$	$\sum_P x^4$	$\sum_Q x^4$
4	$23, 28, 32, 49$	$18, 25, 42, 45$	4,738	7,707,874	7,707,922
8	$21, 24, 45, 51$	$19, 27, 43, 52$	5,643	11,392,083	11,392,179
16	$14, 38, 55, 55$	$3, 43, 54, 54$	7,690	20,424,802	20,424,994
32	$9, 39, 39, 46$	$23, 29, 37, 50$	5,239	9,110,899	9,111,283
64	$17, 28, 44, 59$	$15, 36, 37, 60$	6,490	16,563,634	16,564,402
128	$18, 35, 48, 48$	$26, 29, 44, 52$	6,157	12,222,433	12,223,969
256	$20, 40, 58, 65$	$16, 42, 60, 63$	9,589	31,887,121	31,890,193
512	$57, 57, 63, 64$	$56, 59, 61, 65$	14,563	53,642,179	53,648,323
1,024	$14, 34, 41, 53$	$15, 30, 46, 51$	5,842	12,090,994	12,103,282

For $\mathcal{F}_{51}$ the total reservation uses 33 selected cuts, has maximum distance 51, and respects all capacities. For $\mathcal{F}_{65}$ it uses 53 selected cuts, has maximum distance 65, and respects all capacities. The corresponding assertions are executed before the bridge program processes any interval row.

The family $\mathcal{F}_{100}$ uses the full universal table in Appendix A together with the additional 131,072 pair.

D Selected exact outputs and hashes

The complete interval tables contain 192, 113, and 807 rows, respectively. They are supplied in full rather than typeset here. The key junctions are

Certificate	First relevant interval	Final relevant interval
Lower compressed bridge	[282,204,658, 495,014,629]	[272,308,250,181, 554,860,689,583]
Finite tail junction	[272,308,250,181, 554,860,689,583]	[1,304,696,148,031, 3,868,972,697,957]
Universal family	[1,596,122,112,921, 3,577,546,733,067]	all subsequent overlapping intervals

Table 5: Selected SHA-256 values from the verified manifest.

File	SHA-256
`sw4_certificate.py`	`d85678eb9a4ecb97ea3e0786eb721bd6bb7dc407fbc012775eee76a469675e1d`
`sw4_missing_values.txt`	`c5abe94bd3405d695e42b9e8970f9a95a6ac00e138f0959290cef19ca62c349f`
`sw4_missing_ranges.txt`	`e70a93150c71066fcc6d64e585643b3acf989d0d335a5d90c79486cd3b9f115e`
`sw4_caterpillar_bridge_certificate.py`	`02450eadd42abd1f182a2e2c562f2886e4703a6480bed76a4ef733408230946d`
`sw4_complete_bridge_certificate.cpp`	`05ac482d6cafd8ec999ef2c087f706aae565cfd2b5292e6271724ce111d61aaf`
`sw4_complete_bridge_output.txt`	`549cb209c95154e49c9c50876d3cec58bb332f994321cf44edb390a287664cd3`
`sw4_bridge_independent_check.py`	`883242643f681627e167d78c9c3b54548b786cc75d8b227424f9d20230adcfea`
`sw4_infinite_tail_finite_overlap.cpp`	`9c25622e5253f5ea3396bd66fd88b7885b2f414c1dc06ba3974360523b9e55e9`
`sw4_infinite_tail_finite_overlap.csv`	`c3121d3dc41ebd5bcf16f7fd7fc452abdaac27d025c09d87e5542899f6d9258b`
`sw4_infinite_tail_certificate.py`	`ebe6f805621a58f0284f0de54bbf36b433efe7a0a633c9283a6e9fb34445ee94`
`sw4_infinite_tail_overlaps.csv`	`9c481a07a99b05b44aa062ad7df35be9c5839fcab115359ea37e88264fc33994`
`sw4_infinite_tail_independent_check.py`	`a58da339f5fd7013c130dfaa37d533c5b1263145678cc1152033d9ec892b9ec4`
`sw4_central_seed_witnesses.csv`	`be4b8cf0fd7bf8c7898beb8698174246d53cf1dd3d4ab64c3d774e1836c6ef3b`
`sw4_central_seed_capacity.csv`	`04578eb7611359b92a9af9ca6da6a5ed9548a1feaccb2d7724a5e3b0b9903c47`
`sw4_full_seed_capacity.csv`	`645090a9d9a8b6aab7c03c2116eb1dfb0e53d00b36cf572682b7871738510bc6`

References

[1] Shadow AI Mathematics Project, *Supplementary exact certificates for the Steiner–Wiener 4-index inverse problem*, public reproducibility archive, https://github.com/ShadowUR0/shadow-ai-sw4-candidate-solution, SHA-256 1ed15206d460cac319b0b3683a41ade37db687ecd590a199126b08e189b47342.